

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

30th March 2021

Subject SP4 – Pensions and Other Benefits

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Mark allocations are shown in brackets.*

- Q. 1)** You are the actuarial advisor to the Government of your country, which has a long-established statutory pension system for the organized sector. The design of the system is a pension linked to final pensionable salary (up to a certain threshold) and pensionable service, payable in full from age 60 to all employees in the organized sector who have contributed to the scheme for at least 20 years of their working lifetime. The current maximum amount of the full pension is INR 7,500 per month, and the pensionable salary threshold has historically been increased inline with the increase in the national average earnings index for the country. The country also has other self-managed employer provided retirement schemes which are prevalent in the organized sector.

The pension is funded from contributions from the employers and employees, calculated as a fixed percentage of the pensionable salary (up to a certain threshold). The Government also contributes a fixed percentage to the pension and any shortfall in funding, is met by the Government.

The Government is concerned that the current pension system is becoming increasingly difficult to finance and is unsustainable in the long term in its current form.

- i) Outline the risks to the Government of guaranteeing a pension that is linked to final salary of employees in the organized sector. (3)
- ii) Give reasons why the costs of the current statutory pension system could be increasing. (5)
- iii) Describe ways in which the Government may make changes to the design of the current statutory pension system that can help reduce the cost. (5)
- iv) Comment on some of the main issues that the Government should consider in making the changes you have identified in part (iii). You may consider issues relating to various stakeholders, viz. the Government, employees and employers. (6)

The Labour Minister has proposed that the Government should completely revamp the current pension system. For future service, the proposal is to introduce a National Pension Scheme, which will be a Defined Contribution scheme by design. Under this scheme, the employer and employee will contribute a similar percentage of pensionable salary (as is paid under the current pension system). Contributions would be accumulated in individual accounts to retirement and invested in funds chosen by the individual. At retirement pension benefits would be bought out with an insurance company. To provide more flexibility, employees will have an option to opt out of the new scheme.

- v) Outline the advantages and disadvantages of the suggested proposal. (11)
- [30]**

- Q. 2)** You are the actuary to a reputed listed Company which offers a final salary scheme. For historical reasons, the scheme is unfunded and the liabilities are provided for in the balance sheet on the basis of actuarial valuation. The management of the company is concerned about the size of the unfunded liabilities in the balance sheet and is considering funding the scheme.

- i) Outline the considerations for determining an appropriate investment strategy if the scheme is to be funded. (10)

The HR director has also stated that they are eventually keen to convert the above final salary scheme into a Defined Contribution Scheme. Therefore, for the initial transition period of the next 5 years, the company will pay the higher of the two benefits, as follows:

- Pension based on current final salary scheme and years of service
- Pension equivalent of the value of the member's accumulated contributions (on a cost neutral basis), as determined by the actuary.

- ii) Outline the issues that may be taken into account in determining the rate to use for accumulating the members' contributions and converting them to a pension equivalent. (10)
[20]

Q. 3) A private Security Agency provides securities to the top-notch people of the society. The Agency sponsors a pension scheme which provides final salary benefits to its employees on retirement. The scheme provides benefits to all employees at age 60, including the Bodyguards on their pay-roll, whose earnings are significantly higher than other employees as they are the key to the company's profitability.

The scheme's actuary is about to perform a funding valuation.

- i) Explain the issues the actuary should consider when setting demographic assumptions for the scheme. (4)
- ii) Suggest possible reasons why the actuary should monitor experience relating to these demographic assumptions. (2)
- iii) Suggest, with reasons, examples of demographic experience which may lead to a surplus in the scheme. (3)

The scheme's trustees would prefer the Bodyguards to make other plans for their retirement rather than relying on the pension scheme. They propose that all the Bodyguards transfer their benefits out of the scheme.

- iv) Describe the implications for the pension scheme if the Actuaries leave the scheme and transfer out their existing benefits. (4)
- v) Discuss the advantages and disadvantages for the Bodyguards of moving benefits out of the pension scheme. (6)

One of the points came to the notice of the Scheme's trustees was the review of the investment strategy if the transfer out takes place. They ask further information on what is meant by the same.

- vi) Discuss why a pension scheme would carry out a review of its investment strategy. (5)

The trustee has further asked the Actuary to give his inputs on change in the investment strategy on purchasing annuities by insurance companies on retirement.

- vii) Explain how a scheme's investment policy might be constrained by the need to purchase insurance company annuities on retirement. (4)
[28]

Q. 4) A Company provides the Final Salary Benefits Scheme to its employees. Following the Corona period last year, the Company has suffered major financial losses and hence done a major staff redundancy in certain areas to cover up the losses made in the business during the lockdown.

A bi-yearly valuation of a defined benefit scheme has just been started.

- i) Describe how you would check the data to satisfy yourself that the details for members not yet retired are reliable. (7)

A large surplus has emerged after the valuation exercise, which has arisen from two sources:

- As the sensex has reached all time high, there is a large increase in the market value of the scheme's equity investments during the period since the last valuation, considerably in excess of the assumption used in the previous valuation.
- A reduction in the active membership a year ago post corona times, when the sponsoring employer restructured the business and made a huge staff reduction. The benefits for early leavers in the scheme are such that significant withdrawal surpluses arise.

These two features have had approximately impact in the ratio of 60:40 i.e. 60% comes from the equity investments and 40% comes from the reduction in active membership.

The scheme's actuary has advised that following two changes can be made without detriment to the scheme's long term financial position:

- The employer's contribution rate can be made less from 8% to 5% of scheme salaries for each of the next two years to concentrate and invest more in their core business rather than giving contribution to the pension scheme in these times.
- The rate of pension increases in payment can be changed from a fixed 2.5% per annum to 3.5% or rate of inflation (whichever is less) in order to be perceived well by the employees. This change to the pension increase would apply to pensioners and deferred pensioners as well as current contributing members.

These two changes have the same expected cost, and implementing both of them is affordable.

The employer has proposed that it will implement the contribution reduction, but that there will be no changes to the pension increases. The balance of the surplus will be retained in the scheme.

- ii) Discuss this proposal for distribution of the surplus. (15)
[22]
